

Preran Rai

+91 7025021585 | preranrai3@gmail.com | linkedin.com/in/preran-rai | github.com/PreranRai | [Portfolio](#)

Education

Sahyadri College of Engineering & Management	Sept 2023 – Present
B.E in Computer Science & Engineering (Artificial Intelligence & Machine Learning) – 9.17 GPA	
Kerala State Board of Higher Secondary Education	June 2021 – Mar 2023
Higher Secondary Education – 96.83%	

Research

Multimodal RGB–X-ray Deep Learning for Automated Arecanut Quality Classification (SSRN Preprint)

- Published an SSRN preprint on a multimodal RGB–X-ray deep learning framework for arecanut quality classification, achieving 99.80% accuracy using CNNs with PyTorch and TensorFlow/Keras.

Projects

Tulu-Kalpuga: Tulu Script Learning & Recognition TypeScript, Python, CSS	Link
- Developed an interactive platform for learning Tulu script with a TypeScript-based frontend and clean, accessible UI. - Trained Python-based ML models for character recognition and integrated them with the application workflow. - Designed writing-practice and character-display features to make Tulu script learning intuitive for beginners.	
AgriSeva: Smart Agriculture Assistant FastAPI, React, PyTorch/TensorFlow	Link
- Built a platform for soil-based crop advisory and AI-powered disease detection. - Developed FastAPI ML inference APIs using PyTorch/TensorFlow models. - Designed a simple React interface for farmers to access predictions and recommendations.	
Campus Resource & Timetable Management System JavaScript, SQL, HTML, CSS	Link
- Developed a web-based platform to manage classroom resources, faculty schedules, and automated timetable generation. - Designed relational SQL tables for rooms, faculty, subjects, and schedules with efficient conflict-checking logic. - Built an intuitive JavaScript + HTML/CSS interface for admins to view, update, and publish timetables.	

Technical Skills

Languages: Python, Java, C, C++, JavaScript, SQL.

Libraries & Frameworks: LangChain, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Tensorflow.

Web Technologies: HTML, CSS, React.js.

Databases: MySQL, SQLPlus, MongoDB, PostgreSQL.

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Postman, Microsoft Excel, Google Colab, Vercel, n8n, Figma, Miro, Canva, Docker.

Soft Skills: Communication, Teamwork, Problem-Solving, Adaptability, Time Management.

Certifications

Artificial Intelligence Foundation	Infosys Springboard – Jun 2025
Gained foundational AI knowledge, including agents and search methods.	
Python Foundation	Infosys Springboard – Jun 2025
Built strong basics in Python programming and core syntax.	
Python for Data Science	NPTEL (IIT Madras) – Jan–Feb 2026
Successfully completed a 4-week NPTEL course with Elite certification and secured 73%.	

Achievements

- Finalist at Skill Sangam 2025's 24-hour hackathon, representing Team Zenith among 70 teams.
- Participated in the Nitte HackFest 2026 36-hour hackathon, representing Team CodeInvaders.